NSE Option Chain Analysis Project

Project Summary and Current Status
NIFTY & BANKNIFTY | SQLite + Pandas | Manual NSE CSV Workflow

1. Project Objective

The purpose of this project is to collect NSE option-chain CSV snapshots and build a long-term historical database for NIFTY and BANKNIFTY. The option-chain information is intended to be used as a second confirmation alongside the user's chart analysis. The project is descriptive and analytical; it does not automatically generate buy/sell decisions.

2. Current Architecture

NSE CSV → Python ingestion/analyzer → SQLite database → Pandas analysis → Historical comparison

Layer	Purpose
NSE CSV	Manual download of the latest option-chain data.
Python ingestion	Robustly parses NSE's irregular CSV structure and cleans numeric values.
SQLite	Permanent long-term storage of snapshot-level and strike-level data.
Pandas	Loads historical SQLite data into DataFrames for analysis.
Future analysis	Study changes in OI, Change OI, LTP, Volume and PCR across time and strikes.

3. Project Directory

`/home/manish/trading_project/nse_option_csv/`
• banknifty_option_chain_analyzer_clean.py — NSE CSV ingestion and SQLite storage
• analyze_option_chain_history.py — Pandas historical analysis
• data/banknifty/ — BANKNIFTY CSV files
• data/nifty/ — NIFTY CSV files
• database/nse_option_chain.db — permanent historical database

4. Supported Instruments

• NIFTY
• BANKNIFTY

The ingestion script detects the index from the CSV filename and processes the corresponding data folder.

5. NSE CSV Parsing

The NSE option-chain CSV is not a simple rectangular CSV. It contains a special first row and 23-column data rows with blank outer columns. The parser therefore uses Python's csv reader, selects valid 23-column rows, extracts the useful 21 columns, removes comma formatting from values such as 43,500.00, and converts numeric fields safely. Blank, '-' and NA-like values are treated as missing values.

6. Data Stored Per Strike

The complete strike-level option-chain structure is retained, including:

• CE OI • CE Change OI • CE Volume • CE IV • CE LTP • CE Change • CE Bid Qty • CE Bid • CE Ask • CE Ask Qty • Strike • PE Bid Qty • PE Bid • PE Ask • PE Ask Qty • PE Change • PE LTP • PE IV • PE Volume • PE Change OI • PE OI

7. SQLite Database

Database: `database/nse_option_chain.db`

Table	Purpose
snapshots	One record for each imported CSV snapshot, including time, index, aggregate OI, PCRs, Max Pain and refer
option_chain	All strike-level CE/PE values belonging to each snapshot.

Duplicate protection is implemented using a SHA-256 hash of the CSV content. Re-running the same unchanged file does not create another identical snapshot.

8. Calculated Metrics

The analyzer currently calculates:

• Total CE OI and PE OI • OI PCR = PE OI / CE OI • Total CE Change OI and PE Change OI • Change OI PCR • Total CE Volume and PE Volume • Volume PCR • Maximum CE OI strike • Maximum PE OI strike • Maximum CE Change OI strike • Maximum PE Change OI strike • Max Pain • OI-based reference strike

9. Descriptive Writing / Unwinding Classification

The current script also identifies possible descriptive patterns using OI and LTP changes:

Description	Condition used
Possible Call Writing	CE OI increases + CE LTP decreases
Possible Put Writing	PE OI increases + PE LTP decreases
Possible Call Unwinding	CE OI decreases + CE LTP increases
Possible Put Unwinding	PE OI decreases + PE LTP increases

These are descriptive classifications only, not automatic trading signals.

10. Pandas Historical Analysis

The separate `analyze_option_chain_history.py` script reads the complete historical database into Pandas. It creates date/time fields and compares each snapshot with the previous snapshot for the same index. Once many snapshots are collected, the DataFrame can therefore represent weeks or months of historical option-chain observations.

Current comparison fields include previous snapshot time, previous PCR, previous CE/PE OI, previous CE/PE Change OI, PCR change, CE/PE OI change, CE-PE OI difference and CE-PE Change OI difference.

11. Current Manual Workflow

Step	Action
1	Download the latest NIFTY and/or BANKNIFTY option-chain CSV from NSE when you want to capture a snapshot.
2	Replace the previous CSV in the appropriate data folder.
3	Run the Python ingestion/analyzer script.
4	The new snapshot is inserted into SQLite if the CSV content is new.
5	Run the Pandas analysis script when historical analysis is required.
6	Use the option-chain information as a second confirmation alongside chart analysis.

12. Important Design Decisions

• No fixed snapshot times are assumed. Times such as 10:00 or 14:00 were only examples.
• The user's actual workflow is manual and event-driven: a CSV is captured whenever an interesting chart setup is observed.
• SQLite is the long-term master storage. Pandas is the analysis layer.
• No permanent analysis-output CSV is required.
• The project is not intended to automatically place trades or make buy/sell decisions.

13. Important Caveats

1. The stored snapshot_time is the Python analyzer's execution time in IST. It should be labelled **Run time (IST)**, not treated as an official NSE market timestamp.
2. The reference strike is an **OI-based proxy** calculated from the available option-chain data. It should not be described as actual spot price or guaranteed ATM.
3. Writing/unwinding classifications are descriptive heuristics and should not be interpreted as standalone trading signals.
4. Historical conclusions will become more useful as the number and variety of snapshots increases.

14. Current Status

Core development is complete for the current phase. The database structure, parser, ingestion process, duplicate protection and Pandas history layer are functioning. The recommended action now is to stop adding features and collect historical data over time.

15. Recommended Next Phase

After sufficient history has accumulated (for example, many weeks or months of snapshots), build the next analysis layer around actual research questions. A useful first analysis is strike-level historical tracking:

• Selected strike → CE OI, CE Change OI, CE LTP, CE Volume over time
• Selected strike → PE OI, PE Change OI, PE LTP, PE Volume over time
• OI/PCR changes between snapshots
• Relationship between chart pattern observations and option-chain structure
• Historical case studies rather than automatic prediction

16. Final Project Principle

Keep the data collection simple and reliable first. Build a clean historical dataset before adding advanced analytics. The value of this project will come from comparing real historical option-chain behavior with the chart situations that the user personally observes.